

MW 1.50 RSE Calculation Report – Kai Oi School Steel Frame

Project: Supporting steel frame for 90mm wooden board on assembly hall roof (9.3m span)

1. Introduction

Erection of supporting steel frame for 90 mm wooden board on assembly hall roof. Complies with MW 1.50.

2. Structural Synopsis

EA 50×50×5 mm equal angles, simply supported, hot-dip galvanized.

3. Design Codes

CoP Structural Use of Steel 2011, Dead & Imposed Loads 2011.

4. Material Strength

Grade S275, $p_y = 275 \text{ N/mm}^2$. Weld strength 770 N/mm.

5. Loading (Step-by-Step)

DL (wooden board) = 0.618 kPa. Notional horizontal = 0.5 kPa. ULS: 1.4DL + 1.6LL.

6. Structural Analysis (SAP2000)

Max: $P = 0.84 \text{ kN}$, $V = 1.1 \text{ kN}$, $M = 0.159 \text{ kNm}$.

7. Member Capacity

$V_c = 39.69 \text{ kN} > 1.1 \text{ kN}$. $M_b = 0.6 \text{ kNm} > 0.159 \text{ kNm}$. Interaction = 0.49 < 1.0 O.K.

8. Connections

5mm fillet weld: $163.93 \text{ N/mm}^2 < 770 \text{ N/mm}^2$ O.K. Hilti HST3-M10 anchors (SF=3.0) O.K.

9. Deflection Check

Actual < allowable O.K.

10. Conclusion + RSE Sign

All Adequate.

RSE Signature: _____ Maurice Lee RSE 12/94